

AVENGER FIELD, TX, USA

WMO: 722192

Lat: 32.473N

Lon: 100.466W

Elev: 725

StdP: 92.91

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 23033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-3.9	-16.3	1.0	3.2	-13.7	1.2	4.9	12.9	13.5	12.0	12.9	3.7	10	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.9	19.9	37.0	20.0	35.6	20.2	23.6	30.2	23.0	29.7	22.6	29.2	5.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	18.5	25.0	21.5	17.7	24.8	21.0	17.1	24.8	74.5	30.0	72.3	29.9	70.4	29.1	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.6	9.3	-9.5	39.5	2.6	2.1	-11.4	41.0	-12.9	42.2	-14.4	43.4	-16.2	44.9		
(5)			-10.5	24.5	2.4	0.8	-12.2	25.1	-13.6	25.6	-14.9	26.0	-16.6	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	7.9	9.7	14.5	19.1	22.5	27.1	28.3	28.2	24.6	18.9	13.2	8.3	(6)
(7)		DBStd	8.73	5.53	6.55	5.53	4.60	4.49	2.95	2.59	2.72	3.30	5.04	5.19	5.50	(7)
(8)		HDD10.0	340	106	78	23	2	1	0	0	0	0	5	28	97	(8)
(9)		HDD18.3	1304	323	247	141	45	16	0	0	0	2	55	165	311	(9)
(10)		CDD10.0	3473	41	70	163	276	388	514	567	564	438	282	124	46	(10)
(11)		CDD18.3	1393	1	5	22	69	145	264	308	306	190	73	10	1	(11)
(12)		CDH23.3	15656	15	79	231	757	1523	3121	3778	3637	1801	615	86	13	(12)
(13)	CDH26.7	7709	1	18	61	294	701	1642	2042	1949	792	196	13	1	(13)	
(14)	Wind	WSAvg	4.8	5.0	5.1	5.4	5.9	5.3	5.2	4.2	3.8	3.9	4.6	4.9	4.8	(14)
(15)	Precipitation	PrecAvg	595	24	32	40	48	79	77	44	69	61	55	39	24	(15)
(16)		PrecMax	865	68	121	123	98	189	187	118	196	171	145	120	144	(16)
(17)		PrecMin	310	1	0	1	4	35	4	3	0	3	3	1	0	(17)
(18)		PrecStd	161	19	33	28	32	42	56	34	58	49	40	34	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	29.5	31.3	34.9	37.5	39.8	39.1	39.1	37.0	32.8	28.2	25.2	(19)
(20)			MCWB	11.3	13.1	14.4	15.4	18.1	20.0	19.9	20.3	19.5	18.0	14.9	12.2	(20)
(21)		2%	DB	22.2	25.9	28.5	32.4	35.2	37.6	37.5	37.7	34.6	30.9	25.9	21.6	(21)
(22)			MCWB	10.5	12.1	14.1	15.7	18.5	19.7	20.0	20.2	20.0	18.9	14.8	11.5	(22)
(23)		5%	DB	19.2	22.8	26.2	30.2	32.8	36.1	36.3	36.4	32.9	29.0	23.5	19.1	(23)
(24)			MCWB	9.8	11.5	13.8	15.8	18.5	20.3	20.5	20.4	19.6	18.5	13.9	10.9	(24)
(25)		10%	DB	16.6	19.9	23.6	27.7	30.8	34.1	34.9	35.0	31.3	27.0	21.3	16.9	(25)
(26)			MCWB	8.6	10.8	13.3	15.6	18.6	20.8	20.6	20.6	19.7	17.5	13.2	10.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.8	18.8	20.8	23.1	24.5	24.2	24.0	23.3	22.4	19.1	16.9	(27)
(28)			MCDWB	18.4	20.1	24.3	26.3	28.8	31.3	31.6	30.3	28.5	26.7	22.3	18.9	(28)
(29)		2%	WB	12.9	15.5	17.7	19.9	22.3	23.6	23.4	23.2	22.6	21.4	17.6	14.8	(29)
(30)			MCDWB	18.0	20.5	22.7	25.5	28.2	30.5	30.5	29.9	27.4	26.5	21.1	17.7	(30)
(31)		5%	WB	11.2	13.9	16.7	19.0	21.6	22.9	22.9	22.7	22.1	20.6	16.4	12.8	(31)
(32)			MCDWB	17.0	19.7	21.4	24.5	27.7	30.1	29.8	29.6	26.6	25.7	20.4	17.0	(32)
(33)		10%	WB	9.4	11.9	15.5	17.9	20.8	22.4	22.3	22.2	21.5	19.5	15.1	10.6	(33)
(34)			MCDWB	15.7	18.9	20.8	23.2	27.2	29.5	29.3	29.2	26.4	24.3	19.7	16.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.1	12.7	13.0	13.5	11.9	12.0	11.7	11.6	10.9	12.0	12.0	11.2	(35)
(36)			MCDBR	15.3	16.8	15.8	16.2	14.9	14.4	13.3	13.3	12.8	13.5	13.8	14.7	(36)
(37)			MCWBR	7.5	7.8	6.9	6.7	5.5	4.3	4.0	3.7	3.9	5.6	6.5	7.6	(37)
(38)		5% WB	MCDBR	13.3	14.1	12.1	12.6	12.2	11.6	11.0	11.2	9.9	11.3	11.0	11.9	(38)
(39)			MCWBR	8.0	8.6	6.4	6.4	5.5	4.3	3.8	3.7	3.4	5.9	7.1	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.279	0.291	0.315	0.350	0.368	0.395	0.402	0.391	0.363	0.341	0.308	0.288	(40)
(41)		taud		2.560	2.534	2.475	2.386	2.405	2.362	2.404	2.417	2.432	2.477	2.502	2.539	(41)
(42)		Ebn at Noon		955	973	966	939	919	889	880	888	903	901	908	923	(42)
(43)		Edh at Noon		81	92	105	120	119	124	118	115	109	96	85	78	(43)
(44)	All-Sky Solar Radiation	RadAvg		3.28	4.00	5.23	6.40	6.73	7.27	7.00	6.45	5.37	4.33	3.45	2.89	(44)
(45)		RadStd		0.34	0.50	0.35	0.45	0.63	0.48	0.45	0.40	0.38	0.59	0.39	0.31	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+83

Nomenclature: See separate page